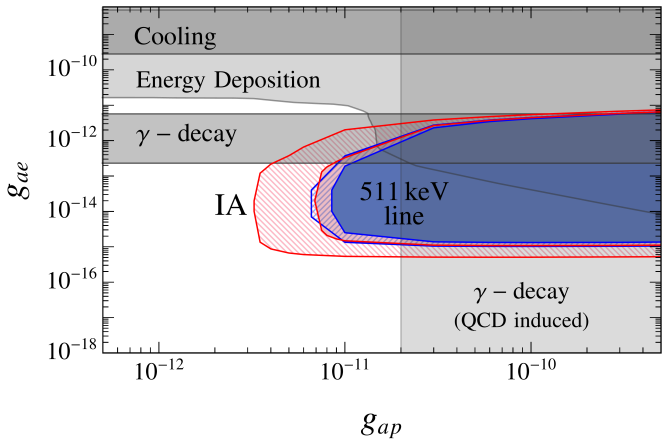

FIG. 10. IA annihilation bounds in the g_{ae} vs g_{ap} plane for an ALP mass $m_a = 100\,\text{MeV}$. We report in red, regions of the parameter space excluded by the arguments related to positron IA, while blue regions are excluded by the non-observation of the 511 keV line. Hatched regions display the uncertainty bands for these limits, ranging from the most conservative to the most optimistic cases discussed in Ref. [24]. We also report in gray, regions of the parameter space excluded by other arguments, namely the SN cooling bound and the loop-induced γ-decay bound on g_{ae} discussed in Ref. [138]. Moreover, we also display constraints from energy deposition in the SN envelope [40], adapted from Ref. [41], and the γ-decay constraint related to the ALP-photon coupling induced by QCD effects, taken from Ref. [44].

IV. DISCUSSION AND CONCLUSION

In this work, we discuss constraints on electrophilic Feebly Interacting Particles (FIPs) produced in Galactic Supernova (SN) explosions during the Milky Way's lifetime. FIPs with masses up to hundreds of MeV are efficiently produced in SN explosions and escape the star, subsequently decaying into electron-positron pairs. Injecting energetic leptons in the Galaxy produces secondary photon fluxes that might be detectable and shed light on the nature and properties of FIPs. Building upon the study conducted in Ref. [24], we discuss the IA signal produced when energetic FIP-produced positrons encounter the free-electrons in the interstellar medium, which produces a peculiar γ-ray signal at tens of MeV. In particular, we analyzed possible signatures related to positron IA induced by the SN emission of axions (coupled to electrons and/or nucleons), sterile neutrinos and dark photons from COMPTEL/EGRET measurements of the diffuse γ-ray background.

We observe that the IA bound is the most constraining astrophysical probe for MeV-scale axions coupled to nucleons and electrons, reaching extremely small electron couplings that cannot be probed otherwise. As already pointed out in Ref. [24], sterile neutrinos mixed with mu and tau flavors can be probed down to $|U_{x4}|^2 \sim 10^{-13}$ for $x = \{\mu, \tau\}$, making IA bounds the strongest ones in the 10-200 MeV mass range. Also in the case of DPs, the IA phenomenology allows one to cover a large portion of the parameter space, providing complementary constraints to cosmological probes.

In conclusion, we show that IA can be considered a novel and powerful γ-ray emission mechanism, making it an important tool for probing new physics and placing stringent constraints over a wide range of positron emitters.

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